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# 2 Stage OTA Design

## Assumptions

$C_L = 10\,pF$     $C_c = 2.2 \times C_L = 2.2\,pF$

unity GBW = 10 MHz     taking $C_c$ to be 3 pF for init. calculation

$V_{OD} = 0.2V$ for all transistors

## Calculations

$g_{m1} = 2\pi \times 10MHz \times C_c = 188.5\,\mu A/V$

$I_{D1} = 18.85\,\mu A \approx 20\,\mu A$

tail current $= 2 \times 20 = 40\,\mu A$

$\left(\frac{W}{L}\right)_{1,2}$: $\quad \frac{W}{L} = \frac{g_{m1}^2}{2(\mu n\, C_{ox} \cdot I_{D1})} = 3.86 \approx 4$

$\left(\frac{W}{L}\right)_{3,4}$: $\quad \frac{W}{L} = \frac{2 \cdot I_{D3}}{\mu p \cdot C_{ox} \cdot (V_{OD})^2} = 10$

$\left(\frac{W}{L}\right)_{0}$: $\quad I_0 = 2 I_{D1}$

$\therefore \left(\frac{W}{L}\right)_0 = 2\left(\frac{W}{L}\right)_1 = 2 \times 4 = 8$

$g_{m5} \approx 2.2\, g_{m1} \left(\frac{C_L}{C_c}\right)$

$\approx 1382.2\,\mu A/V$

taking $g_{m5}$ to be $1500\,\mu A/V$

$I_{D5} = \frac{g_{m5} \cdot (V_{OD})}{2} = \frac{1500 \times 0.2}{2} = 150\,\mu A$

$\left(\frac{W}{L}\right)_5$: $\quad \frac{W}{L} = \frac{g_{m5}^2}{2\,\mu p\, C_{ox} \cdot I_{D5}} = \frac{(1500)^2}{2 \times 100 \times 150} = 75$

$\left(\frac{W}{L}\right)_6$: $\quad \frac{W}{L} = \frac{2\, I_D}{\mu n\, C_{ox}\, (V_{OD})^2} = 32.6 \approx 33$

$R_2 = \frac{1}{g_{m5}} = 667\,\Omega$

$M_1, M_2$ → differential pair (NMOS)

$W = 4\ \mu m$ $I_D = 20\ \mu A$ $r_o = 500\ k\Omega$
$L = 1\ \mu m$
$g_m = 188.5\ \mu A/V$ (200 µA/V when $I_D$ approx. from 18.85 to 20 µ

$M_3\ M_4$ → PMOS current mirror pair

$w = 10\ \mu m$ $I_D = 20\ \mu A$ $r_o = 500\ k\Omega$
$L = 1\ \mu m$
$g_m = 188.5\ \mu A/V$

$M_0$ → tail current source

$w = 8\ \mu m$ $I_D = 40\ \mu A$ $r_o = 250\ k\Omega$
$L = 1\ \mu m$
$g_m \approx 400\ \mu A/V$

$M_5$ → 2nd stage CE-amp

$w = 75\ \mu m$ $I_D = 150\ \mu A$ $r_o = 66.66\ k\Omega$
$L = 1\ \mu m$
$g_{m5} = 1500\ \mu A/V$

$M_6$ → 2nd stage current source

$W = 33\ \mu m$ $I_D = 150\ \mu A$ $r_o = 66.66\ k\Omega$
$L = 1\ \mu m$
$g_m = 1500\ \mu A/V$

(b) other components

ideal curr. source = 4 µA
$M_7$ bias transistor; $W = 0.4\ \mu m$
$L = 1\ \mu m$

$C_c = 3\ pF$
$C_L = 10\ pF$

(c) DC gain

The open loop gain of the amplifier is 67.5 dB
$A_v = 2238.72$

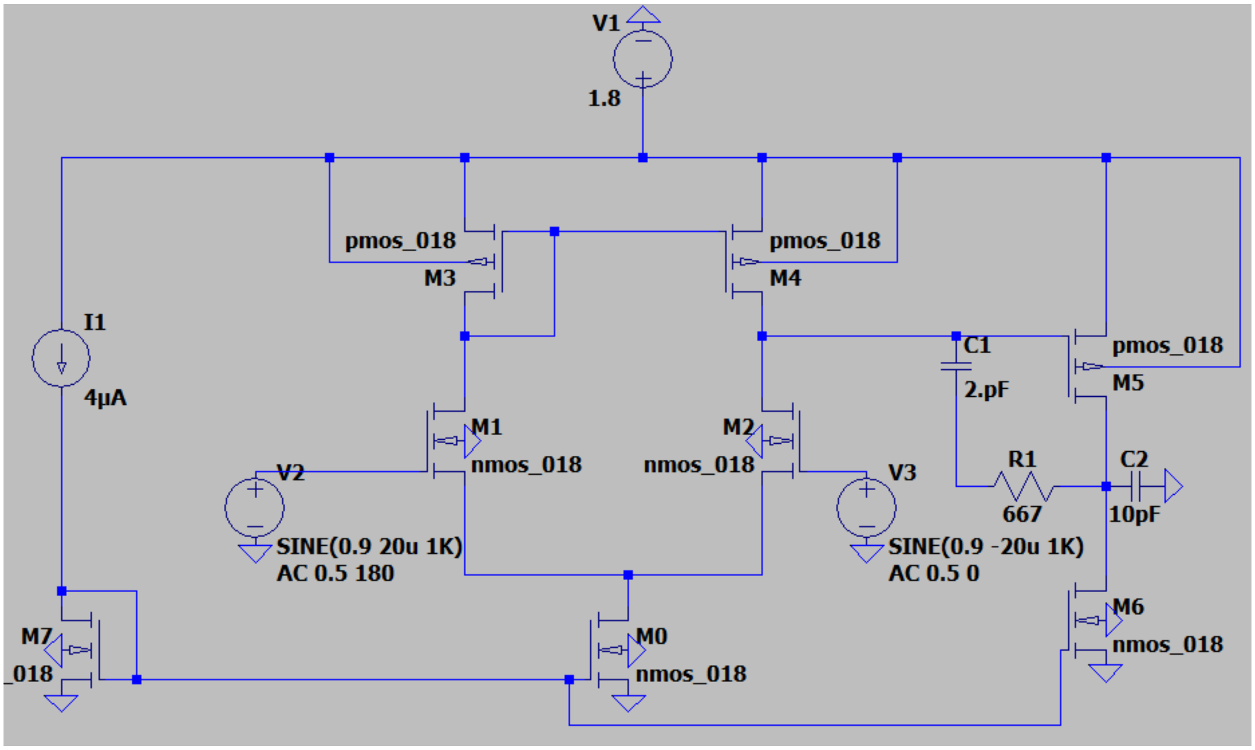
(d) Power consumption

$V_{DD} = 1.8\ V$ ; $I_{net} = I_{tail} + I_{ref} + I_{CE}$
$= 40 + 4 + 150$
$= 194\ \mu A$

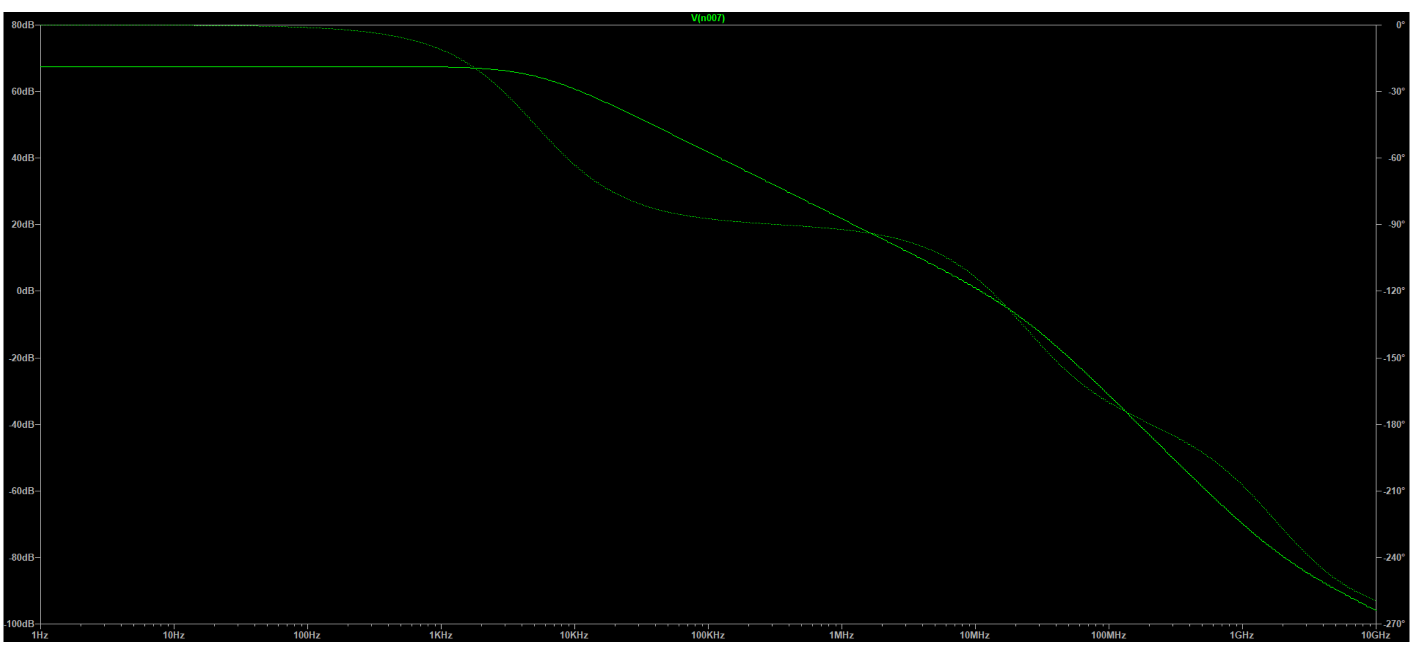
$\boxed{P = 1.8 \times 194 = 349.2\ \mu W}$

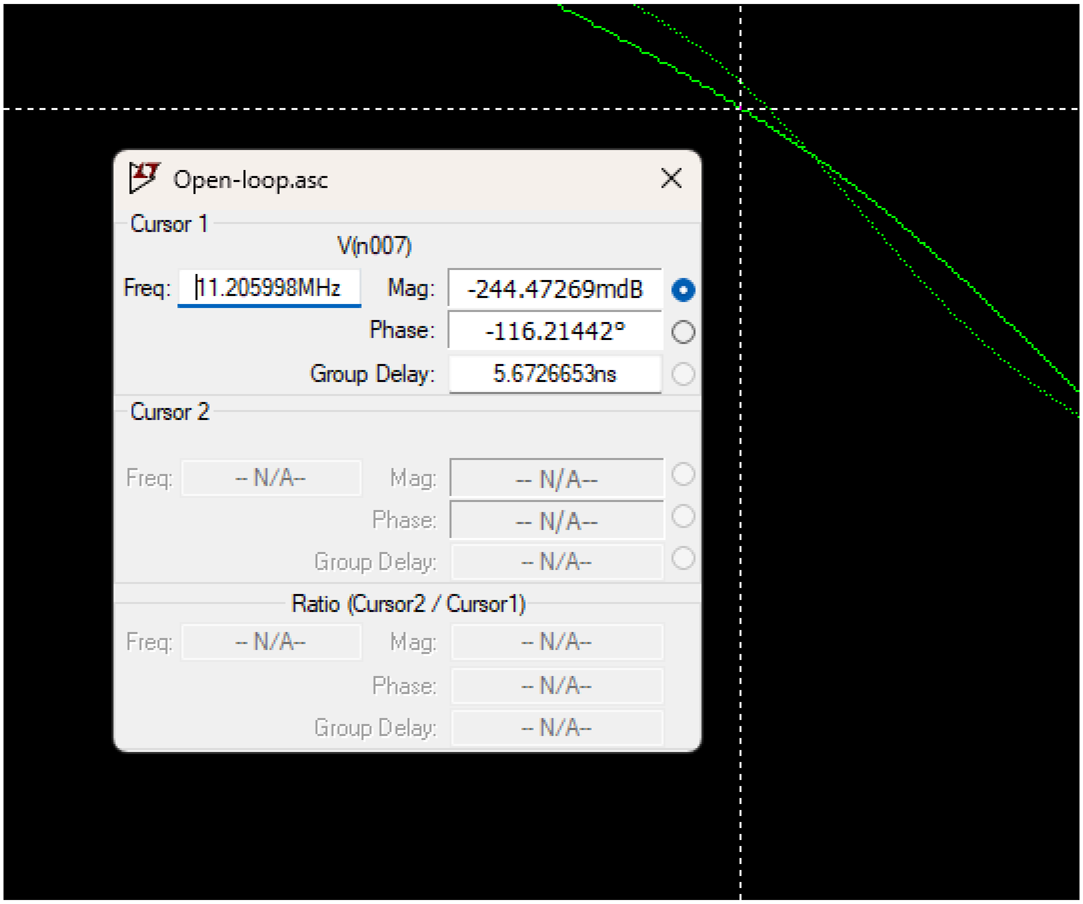
# Open Loop Response

## Schematic:




## Bode Plot:



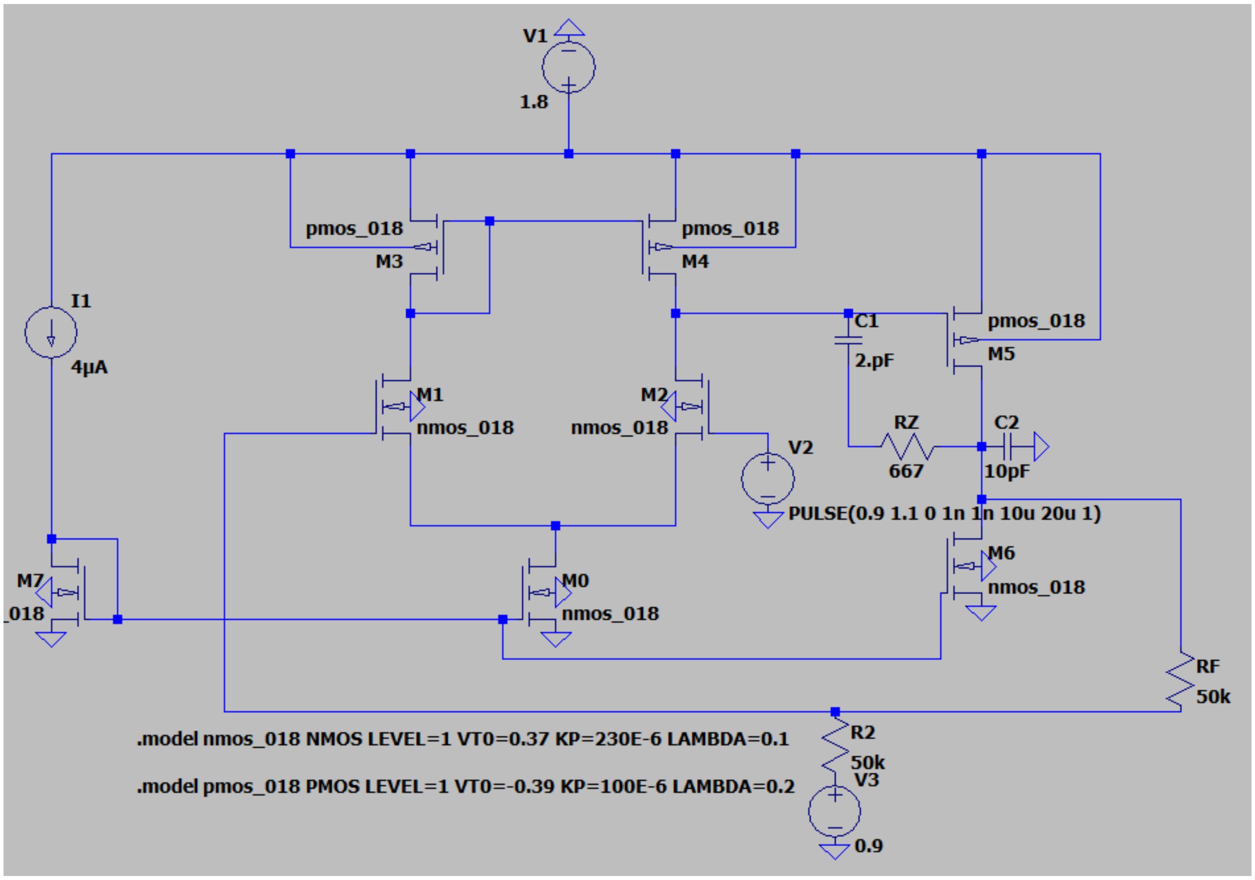



At unity gain, phase margin = -116 + 180 = 64 degree

After plotting the frequency response, the Miller compensation capacitance was tuned from 3pF to 2pF to bring PM closer to 60 degree.

# Closed Loop Response

**Schematic:**




RF= 50k ohm feedback resistor

Closed loop gain = 1 + RF/R1 = 2

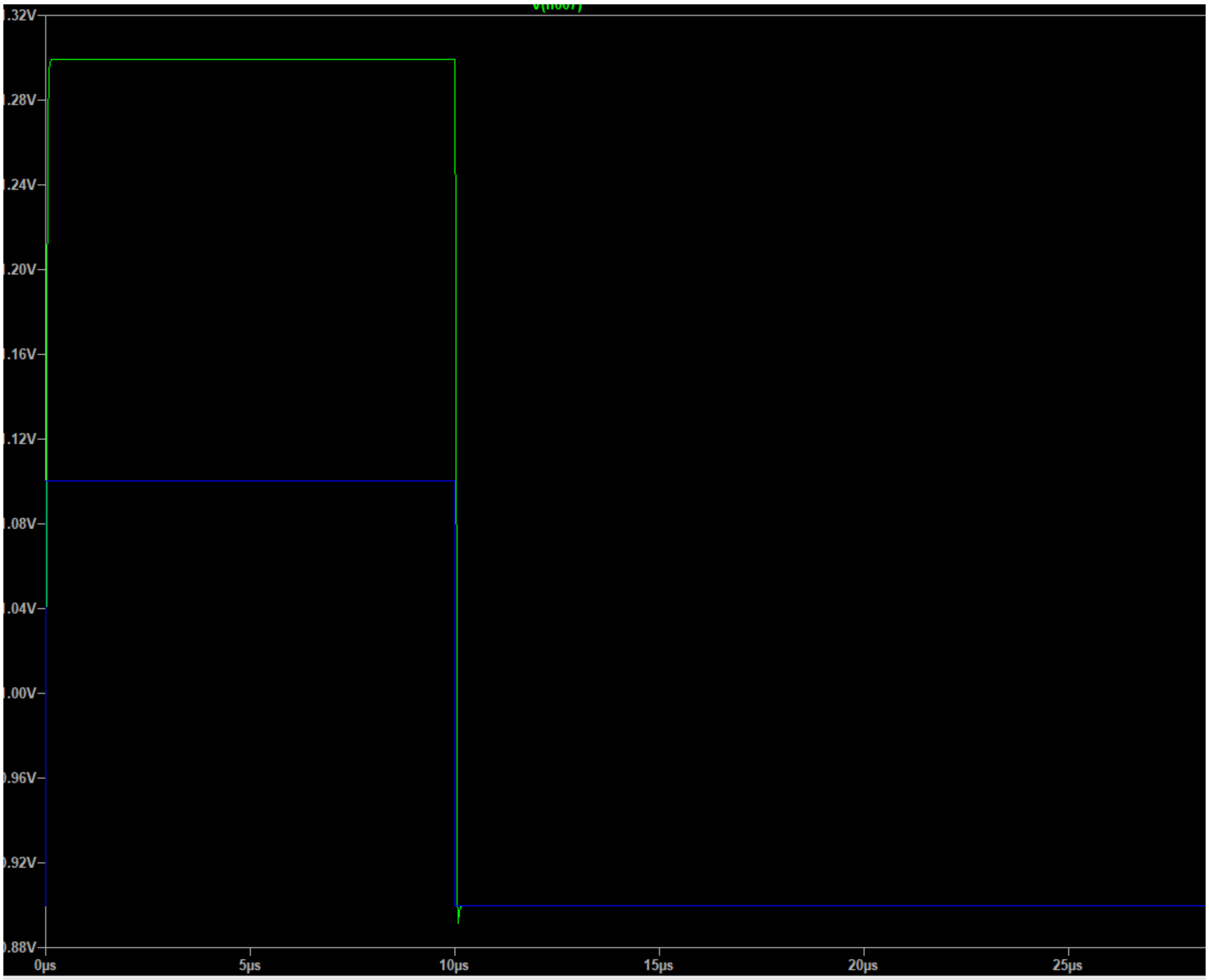
Input pulse is varied from 0.9 to 1.1 V because input voltage needs to be a minimum voltage so as to maintain all transistors in saturation. Vdd/2 = 0.9 taken as DC voltage to be safe.

**Result:**

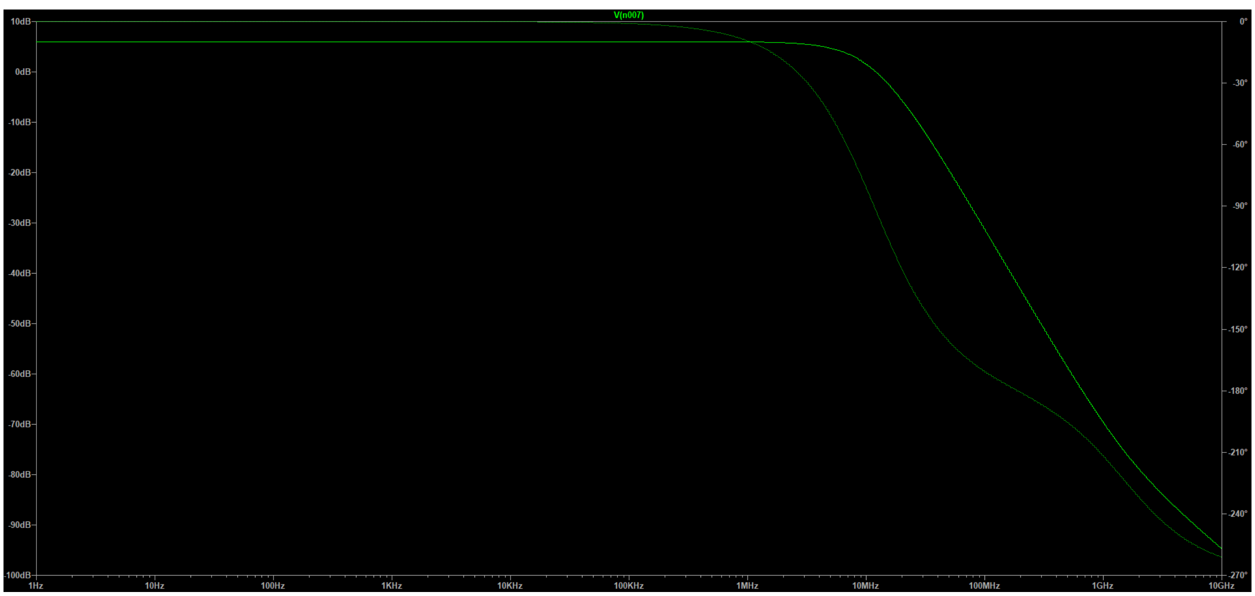
Input pulse = 1.1-0.9 = 0.2 V

Output pulse = 1.299-0.9 = 0.399 V

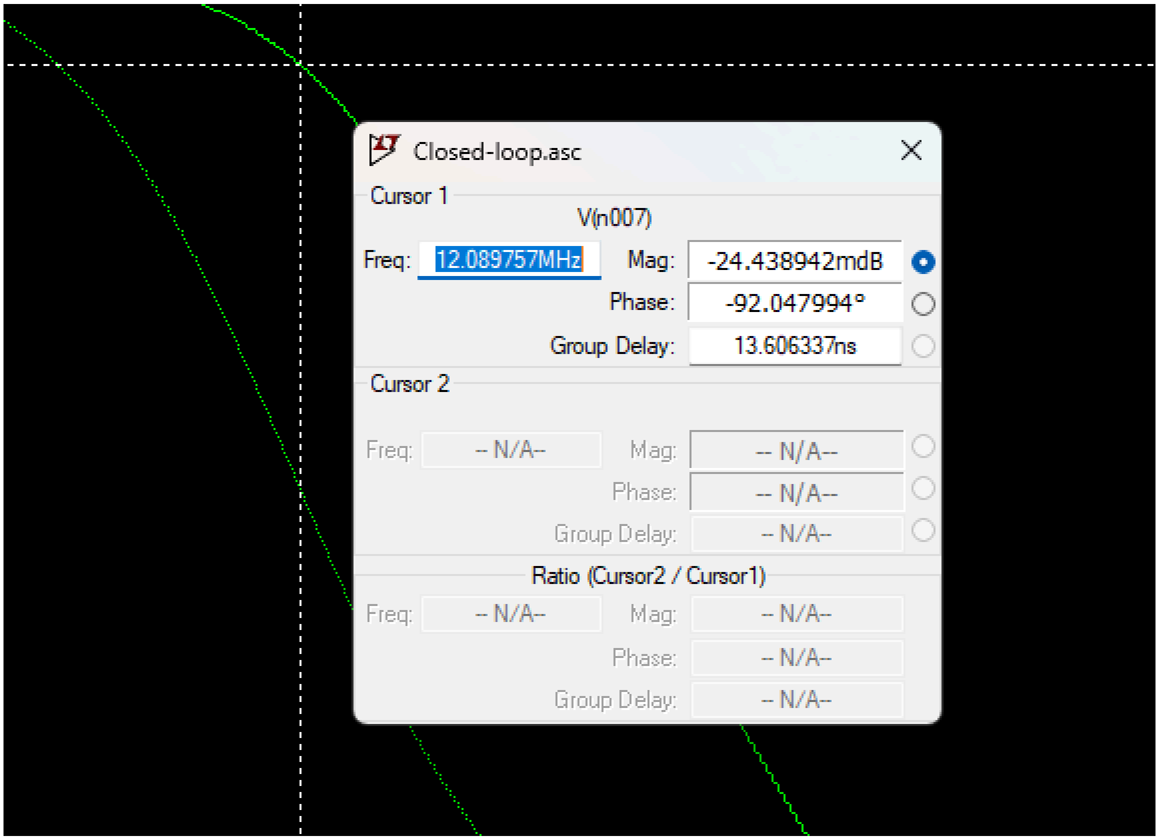
Closed loop gain = 0.399/0.2 = 2




**Frequency Response:**




Closed Loop gain = 6.01 dB




Closed loop phase Margin = -92 + 180 = 88 degree